

Introduction to Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, and decision-making.

What is AI?

AI refers to systems or machines that mimic human intelligence to perform tasks and improve themselves based on the information they collect.

Types of AI

- 1. Narrow AI:** Designed for a specific task.
- 2. General AI:** Can perform any intellectual task like a human.
- 3. Super AI:** Advanced AI beyond human intelligence (theoretical).

Applications of AI

AI is used in healthcare, banking, education, self-driving cars, chatbots, recommendation systems, cybersecurity, and robotics.

Advantages of AI

- Reduces human effort
- Improves accuracy
- Works 24/7
- Handles large amounts of data efficiently

Disadvantages of AI

- High development cost
- Job displacement concerns
- Dependency on machines
- Ethical and privacy issues

Popular AI Technologies

Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Robotics, and Generative AI.

Future of AI

AI is rapidly growing and is expected to transform industries by automating tasks and assisting humans in smarter decision-making.

AI Technologies Overview

AI Technology	Purpose
Machine Learning	Learning from data
Deep Learning	Advanced neural networks
NLP	Understanding human language
Computer Vision	Image and video analysis